

Simple Probability

Probability is the likelihood, or chance, of something happening or not happening.

Probability is a ratio. It compares the number of favorable outcomes to the number of possible outcomes. A favorable outcome is the event you are analyzing.

probability of an event =

number of favorable outcomes

total number of possible outcomes

Probability is often written as a percent.

- A probability of 0% means than an event cannot happen.
- A probability of 100% means than an event is certain to happen.

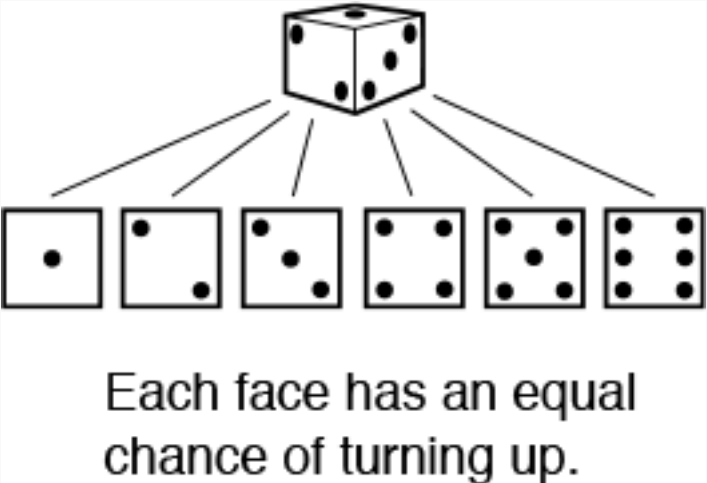
Example 1

[Hide](#)

What is the probability of rolling a 6 using a regular 6-sided die?

Step 1

Count outcomes. Since the die has 6 sides, the total number of possible outcomes is 6. Since there is only 1 side with a 6, the number of favorable outcomes is 1.



Step 2

Write the ratio.

probability of rolling a 6 =

number of favorable outcomes

total number of possible outcomes

=

1

6

The probability of rolling a 6 is $\frac{1}{6}$ or **1 chance in 6** or **about 17%**.

You can also figure out the probability of something *not* happening.

Example 2

[Hide](#)

Oscar has 10 pairs of socks. Of the pairs, 3 are black, 2 are gray, 3 are brown, and 2 are blue. Without looking, Oscar reaches into the drawer and draws out a pair of socks. What is the chance that the socks are *not* blue?

The chance of getting blue is $\frac{2}{10}$, which equals $\frac{1}{5}$, or 20%.

To find the chance of *not* getting blue, subtract from 100%. $100\% - 20\% = \mathbf{80\%}$

Finding the **odds** of an event happening is different from finding probability. To find the odds, write a ratio comparing the number of positive outcomes to the number of negative outcomes. The odds of Oscar drawing out a pair of blue socks are 2 blue to 8 non-blue, or **1 to 4**.

Key Terms

probability

Rate Your Confidence

High

Medium

Low

Continue >>

Category

Lesson 5: Probability

Related Questions:

[Practice questions taken related to this topic](#)